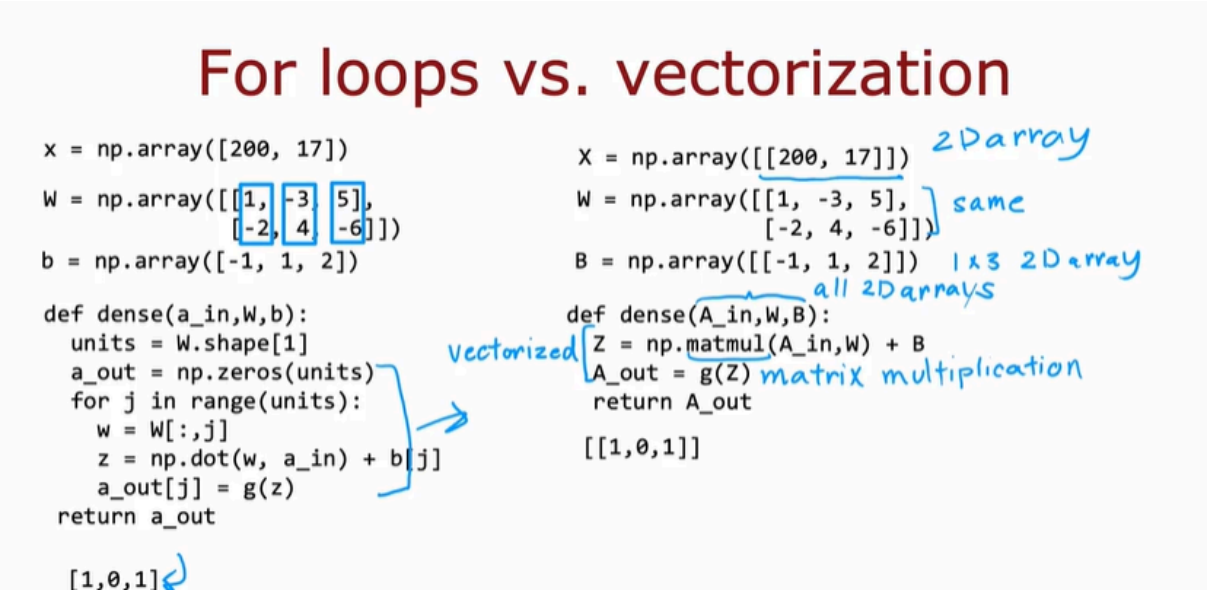
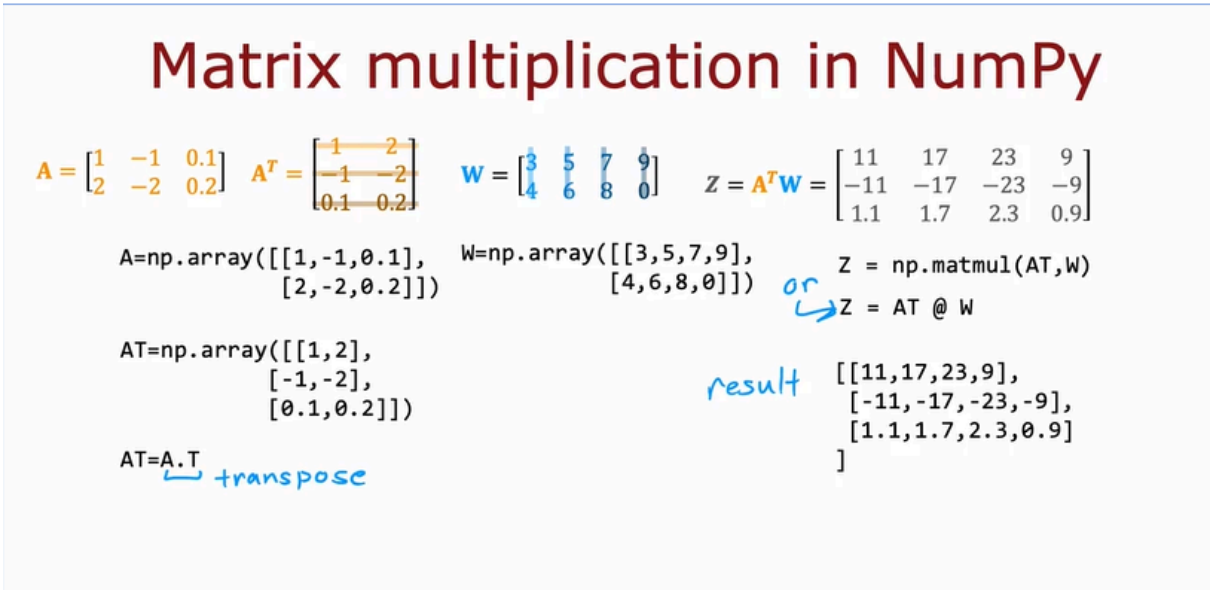
Vectorization

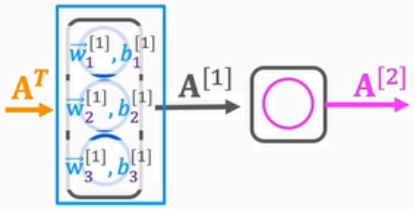
How neural networks are implemented efficiently



Matrix multiplication code



Dense layer vectorized




```
A
AT = np.array([[200, 17]])
W = np.array([[1, -3, 5],
              [-2, 4, -6]])
b = np.array([[-1, 1, 2]])
          a_in
def dense(AT,W,b):
  z = np.matmul(AT,W) + b
  a_out = g(z)   a_in
  return a_out
[[1,0,1]]
```

$A^T = [200 \quad 17]$ (1 × 2)

$W = \begin{bmatrix} 1 & -3 & 5 \\ -2 & 4 & -6 \end{bmatrix}$ (2 × 3)

$B = [-1 \quad 1 \quad 2]$ (1 × 3)

$Z = A^T W + B$

$[165 \quad -531 \quad 900]$

$z_1^{[1]} \quad z_2^{[1]} \quad z_3^{[1]}$

$A = g(Z)$

$[1 \quad 0 \quad 1]$